TEST PLAN

Personal portfolio project

Web shop on Kubernetes

Test plan for release 1.2

Version 1.0

Document history

Version	Date	Author	Comment
0.1	2026-11-07	Jakub Samborski	First draft for release 1.2
1.0	2026-11-07	Jakub Samborski	Open questions answered: free delivery from $100.00, 20 products. Approved.

1 Introduction

1.1 Identification

Item	Value
Test plan ID	TP-WEBSHOP-1.2
System	Web shop on Kubernetes: four Python/Flask microservices and SQL Server
Release	1.2.0 (previous release: 1.1.0)
Author and roles	Jakub Samborski. One-person project: tester, developer and approver.

1.2 Background and purpose

Release 1.1 was tested and released: all 103 test cases passed (see the v1.1 test report). Release 1.2 adds three changes: free delivery to the door for larger orders, a store-style presentation of the products with images, and a larger assortment.

The testing shall verify the new requirements and show, through regression, that nothing that worked in 1.1 is broken.

Goals:

- Every new requirement is verified by at least one test case at the right test level.
- All 103 test cases from 1.1 still pass (regression).
- The full automated test suite passes after every merge, and in a clean environment before release.
- Results and a test report for 1.2 are delivered in docs/v1.2/.

1.3 References

Document	Location
Test report, release 1.1	docs/v1.1/test-report.pdf
Requirements, test cases and results, 1.1	docs/v1.1/test-management.xlsx
Requirements, test cases and results, 1.2	docs/v1.2/test-management.xlsx
Planning and defects	GitHub milestone v1.2: issues and pull requests
How to deploy and run the tests	readme.md

2 Scope

2.1 New requirements

Test cases for the new requirements are written before the code, in docs/v1.2/test-management.xlsx.

ID	Description	Test level	Priority
REQ-42	Delivery to the door is free when the items total $100.00 or more; otherwise it costs $5.00. Pickup at the store is always free.	Unit, E2E	High
REQ-43	The home page shows every product as a card with image, name, price and stock.	E2E	High
REQ-44	The product page shows a larger image and a description.	Unit, E2E	Medium
REQ-45	A product without an image, or whose image cannot be loaded, shows a placeholder.	Unit, E2E	Medium
REQ-46	The assortment is extended to 20 products in total; every product has a description and an image.	Integration, E2E	Medium
REQ-47	Every product image is free to use, and its source is documented in CREDITS.md.	Static	Low

2.2 Regression

All 41 requirements and 103 test cases from 1.1 are run again: 40 unit, 41 integration, 21 E2E and 1 static. Test cases affected by the changes are updated, not removed: for example E2E-008, which counts the products, and checkout tests whose totals reach $100.

2.3 Out of scope

Not tested	Reason
Load and performance	No load environment; unchanged from 1.1.
Browsers other than Chromium	The changes are server-rendered HTML; one browser engine shows the behaviour.
Visual design and image quality	Subjective. Only that the right image or a placeholder is shown is tested.
Filters, search and sorting	Not part of release 1.2.
Account lockout	Postponed to a later release.
Production environment	The system runs locally only.

3 Approach

Testing is requirement-based and risk-based: every requirement has test cases, and the amount of testing follows the risk. The delivery fee concerns money and gets the most attention.

- **Test levels:** as in 1.1. Unit for logic and boundary values, integration for services with the real database, E2E for what the customer sees (Playwright), static for reviews.
- **Techniques:** boundary value analysis for the $100 limit ($99.99, $100.00, $100.01, with door delivery and with pickup); equivalence partitioning for images (present, missing, broken).
- **Automation:** every new test case is automated with pytest, except the static review. Each test carries its test case ID (@pytest.mark.tc).
- **Regression:** the full suite runs on every pull request; python run.py from an empty cluster before the release.
- **Defects:** registered as GitHub issues with severity (Critical, High, Medium, Low), fixed in a branch, and verified by the failing test plus the full regression.

3.1 Iterations

The work is delivered in small pull requests, one at a time. After every merge the shop works and all tests pass.

Step	Content	Done when
1 Test plan	This plan; the 1.2 workbook with the new requirements and test cases (Not run)	Merged
2 Free delivery over $100	Delivery rule, cart and receipt texts, unit and E2E tests	All tests pass
3 Product data	Description and image fields in the database and the product API	All tests pass
4 Product cards	Store-style cards and product page, with placeholder	All tests pass
5 Images and assortment	Images, CREDITS.md, new products	All tests pass
6 Release 1.2	Clean run, results, test report, tag v1.2.0	Exit criteria met

4 Entry and exit criteria

4.1 Entry criteria

- Release 1.1.0 is tagged on GitHub.
- This test plan is approved and merged.
- The new requirements and their test cases are in the 1.2 workbook.
- The system can be deployed with python deploy.py.

4.2 Exit criteria

- All new and all regression test cases are executed.
- No open Critical or High defects.
- Every requirement is covered by at least one passed test case.
- The full automated suite passes in a clean environment.
- The test report for 1.2 is written.

4.3 Suspension and resumption

Testing is paused if the system cannot be deployed or is unstable, or if a Critical defect blocks a main flow (login, cart or checkout). It resumes when the cause is fixed and verified and a deploy succeeds again.

5 Test environment and tools

Component	Version / details
Kubernetes	Docker Desktop, kind cluster, Kubernetes 1.36, one node on a local PC
Services	Python 3.12 containers; SQL Server 2022
Test tools	pytest 8 with requests; Playwright for Python with Chromium
Planning and defects	GitHub: milestone, issues, pull requests and releases

Differences from a production setup: the system runs locally, payment and e-mail are mocked, and there is no HTTPS.

6 Test deliverables

Deliverable	Location
Test plan (this document)	docs/v1.2/test-plan.pdf
Requirements, test cases and results	docs/v1.2/test-management.xlsx
Automated tests	tests/
Results of each test run	reports/test-case-results.csv (generated, not stored in Git)
Defects	GitHub issues
Test report	docs/v1.2/test-report.pdf

7 Risks

Risk	Consequence	Action	Priority
Rounding error at the $100 limit	Wrong delivery fee charged	Boundary value tests with exact decimal amounts	High
Changed product data breaks existing flows	Checkout or orders fail	New fields are optional; full regression on every pull request	High
Images that are not free to use	Legal problem when published	Only CC0 or own images; sources reviewed (REQ-47)	Medium
Tests that depend on the number of products	False failures	Updated in the same pull request as the data change	Medium
Two branches change the workbook at once	Lost changes (binary file)	One branch at a time	Medium
Large images	Slow pages and slower tests	Size limit of about 200 KB per image	Low

8 Open questions

All questions raised while planning are answered. New questions during the testing are registered as GitHub issues.

Question	Owner	Status
Is the free-delivery limit "$100.00 or more" or "more than $100"?	Jakub Samborski	Decided: free delivery from $100.00, i.e. $100.00 or more.
How many products shall the new assortment have?	Jakub Samborski	Decided: 20 products in total.
Does the limit include the delivery fee?	Jakub Samborski	Decided: no, only the items total counts.

9 Approval

Name / role	Decision	Date	Comment
Jakub Samborski, tester	Approved	2026-11-07	Version 1.0